Math 110 — Week 4

Isomorphisms, inverses, products, quotient spaces (Notes Ch. 7–8, LADR 3D–3E, HW4)

§1 Facts used from Weeks 1–3

(W1) A sum $U_1 + \cdots + U_m$ is direct exactly when $u_1 + \cdots + u_m = 0$ with $u_i \in U_i$ forces every $u_i = 0$; then $\dim(U_1 \oplus \cdots \oplus U_m) = \sum \dim U_i$ (Weeks 1–2).

(W2) An independent list extends to a basis. In a space of dimension n, n independent vectors form a basis, and so do n spanning vectors (Week 2).

(W3) Given a basis $v_1, \ldots, v_n$ of V and any $w_1, \ldots, w_n \in W$, there is exactly one linear T with $Tv_j = w_j$. Two linear maps that agree on a basis are equal (Week 3, Prop. 21).

(W4) T is injective $\iff \operatorname{null} T = \{0\}$ (Week 3, Prop. 23).

(W5) Rank–nullity: $\dim V = \dim \operatorname{null} T + \dim \operatorname{range} T$ (Week 3, Prop. 24).

(W6) If $\dim V = \dim W < \infty$: T injective $\iff$ T surjective (Week 3, Prop. 25).

(W7) A nonzero polynomial of degree d has at most d zeros (LADR 4.8). Reason: if $p(a) = 0$, long division gives $p(x) = (x - a)q(x) + r$ with r a constant, and $x = a$ gives $r = 0$; so $p = (x - a)q$ with $\deg q = d - 1$. Any other zero $b \neq a$ of p has $(b - a)q(b) = 0$, so $q(b) = 0$. Induction on d.

(W8) With bases fixed, $T \mapsto \mathcal{M}(T)$ is a linear bijection $\mathcal{L}(V, W) \to \mathbb{F}^{m,n}$, and $\mathcal{M}(ST) = \mathcal{M}(S)\mathcal{M}(T)$ (Week 3, Prop. 26, Def. 20). The rank of a matrix A is the rank of the map $x \mapsto Ax$, the dimension of the span of the columns (Week 3).

§2 This week

§2.1 Isomorphisms and inverses (Ch. 7.1)

The question. When are two vector spaces "the same" as far as linear algebra is concerned, even though their elements look different? For example, is there a linear bijection between $\mathcal{P}_2(\mathbb{R})$ and $\mathbb{R}^3$, and what is its inverse?

Definition (Def. 21 / LADR 3.59, 3.69)**.** A linear map $T\colon V \to W$ is an **isomorphism** (is **invertible**) if there is a linear map $S\colon W \to V$ with $TS = \mathrm{id}_W$ and $ST = \mathrm{id}_V$, where id_V is the identity map $v \mapsto v$ of V (also written I). Then V and W are **isomorphic**, written $V \cong W$. S is a **left inverse** of T if $ST = \mathrm{id}_V$, and a **right inverse** if $TS = \mathrm{id}_W$.

Proposition (Prop. 28 / LADR 3.60: the inverse is unique)
If T is an isomorphism, the map S is unique. It is written T^{-1}.

Proof. 1. Let S_1 and S_2 both satisfy the definition. Then $S_1 T = \mathrm{id}_V$ and $TS_2 = \mathrm{id}_W$.

2. By associativity,
$$S_1 = S_1 \operatorname{id}_W = S_1(TS_2) = (S_1T)S_2 = \operatorname{id}_V S_2 = S_2.$$

□

The computation uses only $S_1T = \mathrm{id}$ and $TS_2 = \mathrm{id}$. So *a left inverse and a right inverse of the same map are equal.* When T has a right inverse, all its left inverses equal that right inverse, and so equal each other; likewise for right inverses when a left inverse exists. With only one side, uniqueness fails: $A = \begin{pmatrix}1 & 2\end{pmatrix} : \mathbb{F}^2 \to \mathbb{F}$ has infinitely many right inverses,
$$\begin{pmatrix}1 & 2\end{pmatrix}\begin{pmatrix}1-2x \\ x\end{pmatrix} = 1 \quad \text{for every } x \in \mathbb{F},$$
and no left inverse: if $SA = I$ and $Av = 0$, then $v = SAv = 0$; but $A(2,-1)^{\mathsf{T}} = 0$.

Proposition (Prop. 29 / LADR 3.63: isomorphism = linear bijection)
A linear map $T\colon V \to W$ is an isomorphism if and only if it is injective and surjective.

The idea

A bijection always has an inverse *function*. The definition asks for a *linear* inverse, so the content is that the inverse function of a linear bijection is automatically linear. To prove $T^{-1}(\lambda w_1 + w_2) = \lambda T^{-1}(w_1) + T^{-1}(w_2)$ without a formula for T^{-1}: both sides are vectors in V, and T is injective, so it is enough to show T sends them to the same vector. Applying T removes every T^{-1}.

Proof. 1. **Assume T is an isomorphism.** If $Tv = 0$ then $v = T^{-1}(Tv) = T^{-1}(0) = 0$: injective. Any w equals $T(T^{-1}w)$: surjective.

2. **Assume T is a bijection.** For $w \in W$ let $T^{-1}(w)$ be the unique v with $Tv = w$. Then $T(T^{-1}w) = w$ and $T^{-1}(Tv) = v$.

3. T^{-1} is linear: by the linearity of T,
$$T\big(T^{-1}(\lambda w_1 + w_2)\big) = \lambda w_1 + w_2 = \lambda T(T^{-1}w_1) + T(T^{-1}w_2) = T\big(\lambda T^{-1}w_1 + T^{-1}w_2\big),$$
and T is injective, so $T^{-1}(\lambda w_1 + w_2) = \lambda T^{-1}w_1 + T^{-1}w_2$.

□

Proposition (Prop. 30 / LADR 3.70: dimension classifies finite-dimensional spaces)
Finite-dimensional V and W are isomorphic if and only if $\dim V = \dim W$.

The idea

An isomorphism changes neither the null space nor the range sizes, so rank–nullity gives equal dimensions. Conversely, send a basis of V to a basis of W; the inverse sends the basis of W back.

Proof. 1. **Assume $T\colon V \to W$ is an isomorphism.** Then $\operatorname{null} T = \{0\}$ and $\operatorname{range} T = W$, so by (W5) $\dim V = 0 + \dim W$.

2. **Assume $\dim V = \dim W = n$.** Choose bases $v_1, \dots, v_n$ of V and $w_1, \dots, w_n$ of W. By (W3) there are linear maps T with $Tv_j = w_j$ and S with $Sw_j = v_j$.

3. $STv_j = v_j$ and $TSw_j = w_j$. So ST and id_V agree on a basis, and TS and id_W agree on a basis. Hence $ST = \mathrm{id}_V$, $TS = \mathrm{id}_W$ (W3).

□

The statement also holds for infinite-dimensional spaces, once "dimension" is made precise as the size of a basis (cited).

Proposition (Prop. 31 / LADR 3.65: with equal dimensions, one condition is enough)
If $\dim V = \dim W < \infty$ and $T \in \mathcal{L}(V, W)$, the following are equivalent: T is an isomorphism; T is injective; T is surjective.

Proof. 1. **An isomorphism is injective and surjective**, by Prop. 29.

2. **Injective or surjective gives both.** Since $\dim V = \dim W < \infty$, injective $\iff$ surjective (W6).

3. **Both give an isomorphism**, by Prop. 29.

□

Proposition (LADR 3D: coordinates are an isomorphism)
If $v_1, \ldots, v_n$ is a basis of V, the map $v \mapsto \mathcal{M}(v)$ from V to $\mathbb{F}^n$ is an isomorphism. With bases of V and W fixed, $T \mapsto \mathcal{M}(T)$ is an isomorphism $\mathcal{L}(V, W) \to \mathbb{F}^{m,n}$.

Proof. 1. $\mathcal{M}$ is linear: if $v = \sum \lambda_j v_j$ and $w = \sum \mu_j v_j$, then $\lambda v + w = \sum(\lambda\lambda_j + \mu_j)v_j$, so $\mathcal{M}(\lambda v + w) = \lambda\mathcal{M}(v) + \mathcal{M}(w)$.

2. $\mathcal{M}$ sends v_j to e_j, so its range contains a basis of $\mathbb{F}^n$ and $\mathcal{M}$ is surjective; $\dim V = n = \dim \mathbb{F}^n$, so it is an isomorphism (Prop. 31).

3. The second statement: $T \mapsto \mathcal{M}(T)$ is a linear bijection (W8), so it is an isomorphism (Prop. 29).

□

So every n-dimensional space is isomorphic to $\mathbb{F}^n$: choosing a basis identifies an abstract n-dimensional space with $\mathbb{F}^n$.

Invertible matrices. A square matrix $A \in \mathbb{F}^{n,n}$ is **invertible** if some B has $AB = BA = I$; then $B = A^{-1}$. Injective, surjective and invertible for A mean the same for the map $x \mapsto Ax$ on $\mathbb{F}^n$; the two notions of invertible agree by the next proposition.

Proposition (LADR 3D: the matrix of the inverse)
Let $\dim V = \dim W = n$, with bases fixed. $T \in \mathcal{L}(V, W)$ is invertible exactly when the square matrix $\mathcal{M}(T)$ is invertible, and then $\mathcal{M}(T^{-1}) = \mathcal{M}(T)^{-1}$.

Proof. 1. **Assume T is invertible.** By (W8), $\mathcal{M}(T)\mathcal{M}(T^{-1}) = \mathcal{M}(TT^{-1}) = \mathcal{M}(\mathrm{id}_W) = I$, and in the same way $\mathcal{M}(T^{-1})\mathcal{M}(T) = \mathcal{M}(\mathrm{id}_V) = I$.

2. **Assume $\mathcal{M}(T)B = B\,\mathcal{M}(T) = I$.** By (W8), $B = \mathcal{M}(S)$ for some $S \in \mathcal{L}(W, V)$. Then $\mathcal{M}(TS) = I = \mathcal{M}(\mathrm{id}_W)$, so $TS = \mathrm{id}_W$, because $\mathcal{M}$ is injective; likewise $ST = \mathrm{id}_V$.

□

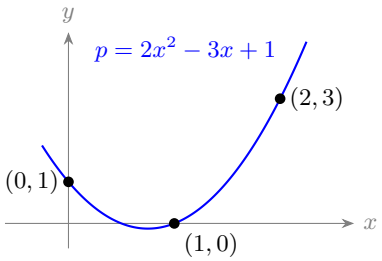
Example 34: interpolation. Given points $(x_0, y_0), \ldots, (x_n, y_n)$ with distinct x_j, there is exactly one $p \in \mathcal{P}_n(\mathbb{R})$ with $p(x_j) = y_j$ for all j.

1. Let $T\colon \mathcal{P}_n(\mathbb{R}) \to \mathbb{R}^{n+1}$, $T(p) = (p(x_0), \ldots, p(x_n))$. Both spaces have dimension $n + 1$.

2. If $Tp = 0$, then p has the $n + 1$ distinct zeros $x_0, \ldots, x_n$. A nonzero polynomial of degree $\leq n$ has at most n zeros (W7), so $p = 0$: T is injective.

3. By Prop. 31, T is bijective: every $(y_0, \ldots, y_n)$ is $T(p)$ for exactly one p.

For the points $(0, 1), (1, 0), (2, 3)$: solving $p(0) = 1$, $p(1) = 0$, $p(2) = 3$ for $p = a + bx + cx^2$ gives $a = 1$, $b + c = -1$, $2b + 4c = 2$, so $c = 2$, $b = -3$, and $p = 2x^2 - 3x + 1$.



Three points with distinct x-values determine exactly one polynomial of degree ≤ 2.

Example 35 (LADR 3.67): existence from injectivity. For every polynomial q there is a polynomial p with $\big((x^2+5x+7)p\big)'' = q$.

1. Choose m with $q \in \mathcal{P}_m(\mathbb{R})$ and let $T\colon \mathcal{P}_m(\mathbb{R}) \to \mathcal{P}_m(\mathbb{R})$, $T(p) = \big((x^2+5x+7)p\big)''$. Multiplying raises the degree by 2 and differentiating twice lowers it by 2, so T maps $\mathcal{P}_m(\mathbb{R})$ to itself.
2. If $Tp = 0$, then $(x^2+5x+7)p$ has degree ≤ 1. For $p \neq 0$ its degree is ≥ 2. So $p = 0$: T is injective.
3. T is an operator on a finite-dimensional space, so it is surjective (Prop. 31). No coefficients were solved for.

Ex. 36: in infinite dimensions this fails. Differentiation $\mathcal{P}(\mathbb{R}) \to \mathcal{P}(\mathbb{R})$ is surjective and not injective.

The 2×2 determinant (computational aside). For $A = \left[\begin{smallmatrix} a & b \\ c & d \end{smallmatrix}\right]$ define $\det A = ad - bc$. This is only a formula for now; determinants in general come at the end of the course.

- Direct multiplication, in both orders:

$$\begin{bmatrix} a & b \\ c & d \end{bmatrix}\begin{bmatrix} d & -b \\ -c & a \end{bmatrix} = \begin{bmatrix} ad-bc & -ab+ba \\ cd-dc & -cb+da \end{bmatrix} = (ad-bc)I,$$

$$\begin{bmatrix} d & -b \\ -c & a \end{bmatrix}\begin{bmatrix} a & b \\ c & d \end{bmatrix} = \begin{bmatrix} da-bc & db-bd \\ -ca+ac & -cb+ad \end{bmatrix} = (ad-bc)I.$$

- If $\det A \neq 0$, then

$$A^{-1} = \frac{1}{ad-bc}\begin{bmatrix} d & -b \\ -c & a \end{bmatrix}.$$

  In words: swap the diagonal entries, negate the other two, divide by $\det A$.
- If $\det A = 0$, then A is not invertible. If $(a,b) \neq (0,0)$, the computation below shows that A is not injective. If $a = b = 0$, every output has first coordinate 0, so A is not surjective.

$$A\begin{bmatrix} b \\ -a \end{bmatrix} = \begin{bmatrix} ab-ba \\ cb-da \end{bmatrix} = 0$$

- Hence A is invertible $\iff$ $\det A \neq 0$.

The question answered. Two spaces are the same for linear algebra when an isomorphism connects them (Def. 21), and two finite-dimensional spaces are isomorphic exactly when their dimensions agree (Prop. 30). For $\mathcal{P}_2(\mathbb{R})$ and $\mathbb{R}^3$: $T(a+bx+cx^2) = (a,b,c)$ sends the basis $1, x, x^2$ to the basis e_1, e_2, e_3, so it is an isomorphism, with inverse $(a,b,c) \mapsto a+bx+cx^2$.

Shortcuts

- **Isomorphism between spaces of equal finite dimension: prove injectivity only.** Reason: Prop. 31. Condition: $\dim V = \dim W < \infty$. Instance: $T(p) = (p(0), p(1), p(2))$ from $\mathcal{P}_2(\mathbb{R})$ to $\mathbb{R}^3$. If $Tp = 0$, then p has three zeros and degree ≤ 2, so $p = 0$; hence T is an isomorphism.
- **Existence without solving.** To show every q has a solution p of a linear equation, show the operator is injective on a finite-dimensional space that contains q. Reason: Prop. 31, as in Ex. 34 and 35. Instance: for every $q \in \mathcal{P}_2(\mathbb{R})$ some $p \in \mathcal{P}_2(\mathbb{R})$ has $p + p' = q$. If $p = a+bx+cx^2$ and $p+p' = (a+b)+(b+2c)x+cx^2 = 0$, then $c = b = a = 0$.

- **To build an isomorphism, send a basis to a basis.** Reason: the proof of Prop. 30 and HW4 P3. Instance: $1 \mapsto (1,1)$, $x \mapsto (1,-1)$ gives the isomorphism $a+bx \mapsto (a+b, a-b)$ from $\mathcal{P}_1(\mathbb{R})$ to $\mathbb{R}^2$.
- **2×2 inverse.** Swap the diagonal entries, negate the other two, divide by $ad-bc$. Reason: HW4 P1. Condition: $ad-bc \neq 0$ in the field you are working in. Instance: the inverse of $\left[\begin{smallmatrix} 2 & 1 \\ 5 & 3 \end{smallmatrix}\right]$ is $\left[\begin{smallmatrix} 3 & -1 \\ -5 & 2 \end{smallmatrix}\right]$.
- **Invertible map $\iff$ invertible matrix.** Reason: the proposition on the matrix of the inverse. Instance: $T(a,b) = (a+b, a-b)$ has $\mathcal{M}(T) = \left[\begin{smallmatrix} 1 & 1 \\ 1 & -1 \end{smallmatrix}\right]$ with $\det = -2$, so $T^{-1}(x,y) = (\frac{x+y}{2}, \frac{x-y}{2})$.

Traps

- In infinite dimensions, injective does not give surjective: the right shift $R(a_1, a_2, \dots) = (0, a_1, a_2, \dots)$ is injective, but $(1,0,0,\dots)$ is not in its range. Surjective does not give injective either: differentiation on $\mathcal{P}(\mathbb{R})$ (Ex. 36).
- Whether $ad-bc \neq 0$ depends on the field: $\left[\begin{smallmatrix} 1 & 2 \\ 3 & 1 \end{smallmatrix}\right]$ has $ad-bc = -5$, so it is invertible over $\mathbb{R}$ but not over $\mathbb{F}_5$.
- A one-sided inverse need not be unique, and need not be an inverse: $\begin{pmatrix} 1 & 2 \end{pmatrix}$ has the right inverses $(1-2x, x)^{\mathsf{T}}$ for every x, and none of them is a left inverse, since a 2×1 matrix times a 1×2 matrix has rank at most 1 and cannot be I_2.
- A non-square matrix is never invertible: $\mathbb{F}^n \cong \mathbb{F}^m$ only when $n = m$ (Prop. 30).
- "Isomorphic" says that an isomorphism exists, not which one: $a+bx+cx^2 \mapsto (a,b,c)$ and $p \mapsto (p(0), p(1), p(2))$ are two different isomorphisms $\mathcal{P}_2(\mathbb{R}) \to \mathbb{R}^3$.

§2.2 Products of maps and change of basis (Ch. 7.3, LADR 3D)

The question. On $\mathcal{P}(\mathbb{R})$ let $S(p) = p'$ and $T(p)(x) = \int_0^x p(t)\,dt$. Check that $S(T(p)) = p$ for every p. Is $T(S(p)) = p$ for every p? For square matrices, does $AB = I$ force $BA = I$?

Proposition (Prop. 32 / LADR 3D: inverses and products)

Let $S, T \in \mathcal{L}(V)$.

(i) If S and T are invertible, so is ST, and $(ST)^{-1} = T^{-1}S^{-1}$.

(ii) If ST is invertible, then S is surjective and T is injective. If V is finite-dimensional, then S, T and TS are all invertible.

The idea

In ST, T acts first and S acts last. If T sent two vectors to the same vector, so would ST; so ST injective forces T injective. Every output of ST is an output of S; so ST surjective forces S surjective. Neither argument says anything about the other map, and in infinite dimensions the other half can fail (the examples after Prop. 33). In finite dimensions, Prop. 31 then gives that S and T are invertible.

Proof. 1. **(i) $T^{-1}S^{-1}$ is the inverse of ST.** Multiply in both orders:

$$(T^{-1}S^{-1})(ST) = T^{-1}(S^{-1}S)T = T^{-1}T = I,$$
$$(ST)(T^{-1}S^{-1}) = S(TT^{-1})S^{-1} = SS^{-1} = I.$$

2. **S is surjective.** Any v is

$$v = (ST)(ST)^{-1}v = S\big(T(ST)^{-1}v\big) \in \operatorname{range} S.$$

3. **T is injective.** If $Tv = 0$, then $v = (ST)^{-1}(ST)v = (ST)^{-1}S(0) = 0$.

4. If $\dim V < \infty$, Prop. 31 makes S and T invertible, and (i) makes TS invertible.

□

Proposition (Prop. 33 / LADR 3D.11: one side is enough in finite dimensions)
If V is finite-dimensional and $S, T \in \mathcal{L}(V)$, then $ST = I \iff TS = I$.

Proof. 1. **Assume $ST = I$.** Then ST is invertible, so T is invertible (Prop. 32). Then $S = S(TT^{-1}) = (ST)T^{-1} = T^{-1}$, so $TS = TT^{-1} = I$.

2. **Assume $TS = I$.** By Prop. 32(ii) applied to the product TS, S is injective and T is surjective, so both are invertible (Prop. 31). Then $T = T(SS^{-1}) = (TS)S^{-1} = S^{-1}$, and $ST = SS^{-1} = I$.

□

Example 38: $ST = I$ but $TS \neq I$ (here $V = \mathcal{P}(\mathbb{R})$ is infinite-dimensional). On $\mathcal{P}(\mathbb{R})$ let $S(p) = p'$ and $T(p)(x) = \int_0^x p(t)\,dt$. Then $ST(p) = p$ by the fundamental theorem of calculus, so $ST = I$. But $TS(p) = \int_0^x p'(t)\,dt = p - p(0)$, and $TS(1) = 0 \neq 1$.

Example 37: ST invertible, but neither S nor T (again $V = \mathcal{P}(\mathbb{R})$). On $\mathcal{P}(\mathbb{R})$ let $S(p) = p'$ and $T(p)(x) = \int_0^{x+1} p(t)\,dt$.

1. By the fundamental theorem of calculus,
$$ST(p)(x) = \frac{d}{dx}\int_0^{x+1} p(t)\,dt = p(x+1).$$
This map has the inverse $p(x) \mapsto p(x-1)$.
2. S is not injective: $S(1) = 0$.
3. $\operatorname{range} T = U = \{q : q(-1) = 0\}$. Each $T(p)$ vanishes at $x = -1$, since $\int_0^0 = 0$. Conversely, if $q(-1) = 0$, let $p(x) = q'(x-1)$; then
$$T(p)(x) = \int_0^{x+1} q'(t-1)\,dt = q(x) - q(-1) = q(x).$$
So T is not surjective: $1 \notin U$.
4. $TS(c) = T(0) = 0$ for every constant c, so TS is not injective.
5. The inverses in the formula below are $T^{-1}(q)(x) = q'(x-1)$ on U (step 3) and $(S|_U)^{-1}(p)(x) = \int_{-1}^x p(t)\,dt$, which lies in U and has derivative p.

The inverse formula still holds in a modified form. Suppose ST is invertible, with V possibly infinite-dimensional. T is injective (Prop. 32), so T is a bijection from V onto $\operatorname{range} T$, and its inverse $T^{-1}\colon \operatorname{range} T \to V$ is linear (Prop. 29). Then $S|_{\operatorname{range} T} = (ST)T^{-1}$, because both sides send Tv to STv. It is a composition of two invertible maps, so it is invertible, and by Prop. 32(i)
$$(ST)^{-1} = \big((S|_{\operatorname{range} T})\,T\big)^{-1} = T^{-1}(S|_{\operatorname{range} T})^{-1}.$$

The question answered. On $\mathcal{P}(\mathbb{R})$, $ST = I$ but not $TS = I$: $TS(1) = T(0) = 0 \neq 1$ (Ex. 38). For operators on a finite-dimensional space the answer is yes (Prop. 33), so for square matrices $AB = I$ gives $BA = I$.

Proposition (LADR 3D: change of basis)
Let $T \in \mathcal{L}(V)$ and let B, B' be bases of V. Let P be the matrix whose column j holds the B-coordinates of the j-th vector of B'. Then P is invertible and
$$\mathcal{M}_{B'}(T) = P^{-1}\,\mathcal{M}_B(T)\,P.$$

Proof. 1. $P = \mathcal{M}^{B}_{B'}(\mathrm{id})$, the matrix of the identity map with B' in the domain and B in the target: column j is the B-coordinates of $\mathrm{id}(b'_j) = b'_j$. Likewise $\mathcal{M}^{B'}_{B}(\mathrm{id})\,P = \mathcal{M}^{B'}_{B'}(\mathrm{id} \circ \mathrm{id}) = I$ and $P\,\mathcal{M}^{B'}_{B}(\mathrm{id}) = I$, so $P^{-1} = \mathcal{M}^{B'}_{B}(\mathrm{id})$.

2. $T = \mathrm{id} \circ T \circ \mathrm{id}$, read with the bases $B' \to B \to B \to B'$. By (W8),

$$\mathcal{M}_{B'}(T) = \mathcal{M}_B^{B'}(\mathrm{id})\,\mathcal{M}_B(T)\,\mathcal{M}_{B'}^{B}(\mathrm{id}) = P^{-1}\mathcal{M}_B(T)P.$$

□

Worked example: N in the basis $(1,1),(1,-1)$. For $N(x,y) = (y,0)$, B standard and $B' = ((1,1),(1,-1))$:

$$P = \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}, \qquad P^{-1}\mathcal{M}(N)P = \frac{1}{2}\begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}\begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}\begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix} = \frac{1}{2}\begin{bmatrix} 1 & -1 \\ 1 & -1 \end{bmatrix}.$$

Check directly: $N(1,1) = (1,0) = \frac{1}{2}(1,1) + \frac{1}{2}(1,-1)$, so the first column is $(\frac{1}{2},\frac{1}{2})$; $N(1,-1) = (-1,0) = -\frac{1}{2}(1,1) - \frac{1}{2}(1,-1)$, so the second column is $(-\frac{1}{2},-\frac{1}{2})$.

Shortcuts

- **Check one side only.** For operators on a finite-dimensional space, and so for square matrices, $ST = I$ gives $TS = I$ and $S = T^{-1}$. Reason: Prop. 33. Instance: $\left[\begin{smallmatrix} 1 & 1 \\ 0 & 1 \end{smallmatrix}\right]\left[\begin{smallmatrix} 1 & -1 \\ 0 & 1 \end{smallmatrix}\right] = I$, so the product in the other order is I too, with no further multiplication.
- **Inverse of a product: reverse the order.** $(ST)^{-1} = T^{-1}S^{-1}$. Reason: Prop. 32(i). Instance: for $S = \left[\begin{smallmatrix} 2 & 0 \\ 0 & 1 \end{smallmatrix}\right]$ and $T = \left[\begin{smallmatrix} 1 & 1 \\ 0 & 1 \end{smallmatrix}\right]$,

$$(ST)^{-1} = T^{-1}S^{-1} = \begin{bmatrix} 1 & -1 \\ 0 & 1 \end{bmatrix}\begin{bmatrix} 1/2 & 0 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} 1/2 & -1 \\ 0 & 1 \end{bmatrix}.$$

- **From ST invertible: T injective, S surjective**, in any dimension. Reason: Prop. 32(ii). Instance: if $T\colon V \to W$ and $S\colon W \to V$ satisfy $ST = \mathrm{id}_V$, then T is injective, so $\dim V \le \dim W$.
- **One-sided inverses in finite dimensions.** $T \in \mathcal{L}(V,W)$ has a left inverse exactly when T is injective, and a right inverse exactly when T is surjective. Reason: the "only if" parts are the arguments of Prop. 32(ii). For a left inverse, use R1. For a right inverse, pick v_i with $Tv_i = w_i$ for a basis w_i of W and set $S(w_i) = v_i$; then $TS = \mathrm{id}_W$ on a basis. Instance: $\begin{pmatrix} 1 & 2 \end{pmatrix}$ is surjective; $Tv = 1$ for $v = (1,0)$, giving the right inverse $(1,0)^{\mathsf{T}}$.
- **Change of basis.** $\mathcal{M}_{B'}(T) = P^{-1}\mathcal{M}_B(T)P$ with the new basis vectors as the columns of P. Reason: the proposition above. Instance: the worked example for N.

Traps

- In infinite dimensions $ST = I$ does not give $TS = I$: in Ex. 38, $TS(1) = 0$.
- In infinite dimensions ST invertible does not make S or T invertible: in Ex. 37, $S(1) = 0$ and $1 \notin \operatorname{range} T$, since 1 is not 0 at -1.
- $(ST)^{-1} \neq S^{-1}T^{-1}$ in general: with S, T above, $S^{-1}T^{-1} = \left[\begin{smallmatrix} 1/2 & -1/2 \\ 0 & 1 \end{smallmatrix}\right] \neq \left[\begin{smallmatrix} 1/2 & -1 \\ 0 & 1 \end{smallmatrix}\right]$.
- In the change-of-basis formula, P has the *new* basis vectors (in old coordinates) as columns, and P^{-1} is on the left. For N and the new basis $(1,1),(0,1)$, $P = \left[\begin{smallmatrix} 1 & 0 \\ 1 & 1 \end{smallmatrix}\right]$ and $P^{-1}\mathcal{M}(N)P = \left[\begin{smallmatrix} 1 & 1 \\ -1 & -1 \end{smallmatrix}\right]$, which is correct: $N(1,1) = (1,0) = (1,1) - (0,1)$. The reversed product $P\,\mathcal{M}(N)P^{-1} = \left[\begin{smallmatrix} -1 & 1 \\ -1 & 1 \end{smallmatrix}\right]$ is wrong.

§2.3 Products (Ch. 8.1)

The question. $\mathbb{R}^2 \times \mathbb{R}^3$ is the set of pairs $\big((x_1,x_2),(y_1,y_2,y_3)\big)$. What is its dimension, what is a basis, and how is it related to $\mathbb{R}^5$?

Products. The **product** of vector spaces $V_1, \dots, V_m$ is the set of lists

$$V_1 \times \cdots \times V_m = \{(v_1, \dots, v_m) : v_i \in V_i\}.$$

Proposition (Prop. 34 / LADR 3.87, 3.89: the product is a vector space)
With the componentwise operations below, $V_1 \times \cdots \times V_m$ is a vector space.

$$(v_1, \dots, v_m) + (w_1, \dots, w_m) = (v_1 + w_1, \dots, v_m + w_m), \qquad \lambda(v_1, \dots, v_m) = (\lambda v_1, \dots, \lambda v_m)$$

Proof. 1. Each axiom holds in every slot, because it holds in each V_i. For example, the commutative law for $+$ in slot i is $v_i + w_i = w_i + v_i$ in V_i.

2. The zero is $(0, \dots, 0)$, and the additive inverse of $(v_1, \dots, v_m)$ is $(-v_1, \dots, -v_m)$, since their sum is $(0, \dots, 0)$.

□

- Ex. 39, 40: $\mathbb{R}^n = \mathbb{R} \times \cdots \times \mathbb{R}$, and $\mathbb{R}^m \times \mathbb{R}^n \cong \mathbb{R}^{m+n}$ by the isomorphism

$$((x_1, \dots, x_m), (y_1, \dots, y_n)) \mapsto (x_1, \dots, x_m, y_1, \dots, y_n).$$

- The spaces $V_1 \times (V_2 \times V_3)$, $(V_1 \times V_2) \times V_3$ and $V_1 \times V_2 \times V_3$ are different sets, but $(v_1, (v_2, v_3)) \mapsto ((v_1, v_2), v_3) \mapsto (v_1, v_2, v_3)$ are isomorphisms, and $(v, w) \mapsto (w, v)$ is an isomorphism $V \times W \to W \times V$, so they are usually not distinguished.
- Let $V_{(i)} = \{(0, \dots, 0, v, 0, \dots, 0) : v \in V_i\}$ (v in slot i). The map $v \mapsto (0, \dots, 0, v, 0, \dots, 0)$ is linear, injective (the only v sent to 0 is 0) and onto $V_{(i)}$, so it is an isomorphism $V_i \to V_{(i)}$ (Prop. 29).

Proposition (Prop. 35, 36 / LADR 3.92: the product is the direct sum of the $V_{(i)}$)
$V_1 \times \cdots \times V_m = V_{(1)} \oplus \cdots \oplus V_{(m)}$. If the V_i are finite-dimensional, $\dim(V_1 \times \cdots \times V_m) = \dim V_1 + \cdots + \dim V_m$.

Proof. 1. $(v_1, \dots, v_m) = (v_1, 0, \dots, 0) + \cdots + (0, \dots, 0, v_m)$, and this is the only way to write it as a sum with one term from each $V_{(i)}$, because the term from $V_{(i)}$ must have v_i in slot i.

2. By (W1), the dimension is $\sum \dim V_{(i)}$, and $\dim V_{(i)} = \dim V_i$ by Prop. 30, since $V_{(i)} \cong V_i$.

□

- Concretely: if $v_1, \dots, v_n$ is a basis of V and $w_1, \dots, w_m$ a basis of W, then $(v_1, 0), \dots, (v_n, 0), (0, w_1), \dots, (0, w_m)$ is a basis of $V \times W$ (HW4 P5).
- The dimension of a product is the *sum* of the dimensions. ($\mathcal{L}(V, W)$ is a space whose dimension is $\dim V \cdot \dim W$; Week 3, Prop. 26.)
- The product is also called the *external direct sum*. Some books write $V \oplus W$ for $V \times W$; this course does not.

Proposition (Prop. 37 / LADR 3.93: products and sums)
For subspaces $U_1, \dots, U_m$ of V, the map $\Gamma\colon U_1 \times \cdots \times U_m \to U_1 + \cdots + U_m$, $\Gamma(u_1, \dots, u_m) = u_1 + \cdots + u_m$, is linear and surjective. It is injective exactly when the sum is direct.

Proof. 1. Γ **is linear**, since addition in the product is componentwise:

$$\Gamma\big(\lambda(u_1, \dots, u_m) + (u_1', \dots, u_m')\big) = \sum_i (\lambda u_i + u_i') = \lambda\,\Gamma(u_1, \dots, u_m) + \Gamma(u_1', \dots, u_m').$$

2. Γ **is surjective**: every element of $U_1 + \cdots + U_m$ is $u_1 + \cdots + u_m$ with $u_i \in U_i$, by definition.

3. Γ **is injective exactly when the sum is direct.** $\operatorname{null}\Gamma = \{0\}$ says that $u_1 + \cdots + u_m = 0$ forces every $u_i = 0$, which is the criterion (W1).

□

Rank–nullity for Γ gives

$$\dim(U_1 + \cdots + U_m) = \sum \dim U_i - \dim \operatorname{null} \Gamma \leq \sum \dim U_i,$$

with equality exactly when the sum is direct. **A sum is direct $\iff$ its dimension is the sum of the dimensions.**

The question answered. The dimension is $2 + 3 = 5$ (Prop. 36). A basis is $(e_1, 0), (e_2, 0), (0, e_1), (0, e_2), (0, e_3)$ (HW4 P5). The isomorphism of Ex. 40 identifies the space with $\mathbb{R}^5$.

Shortcuts

- **Basis of a product.** Use $(v_i, 0)$ and $(0, w_j)$. Reason: HW4 P5. Instance: $\mathbb{R}^2 \times \mathcal{P}_1(\mathbb{R})$ has the basis $(e_1, 0), (e_2, 0), (0, 1), (0, x)$, so its dimension is 4.
- **Is a sum direct? Compare dimensions.** $U_1 + \cdots + U_m$ is direct exactly when $\dim(U_1 + \cdots + U_m) = \sum \dim U_i$. Reason: rank–nullity for Γ (Prop. 37). Instance (Practice Midterm 1, Q1): for $U = \operatorname{span}((1,1,0,0),(0,1,1,0))$ and $W = \operatorname{span}((0,0,1,1),(1,0,0,-1))$, the four vectors are independent, so $\dim(U + W) = 4 = 2 + 2$ and $\mathbb{R}^4 = U \oplus W$.

Traps

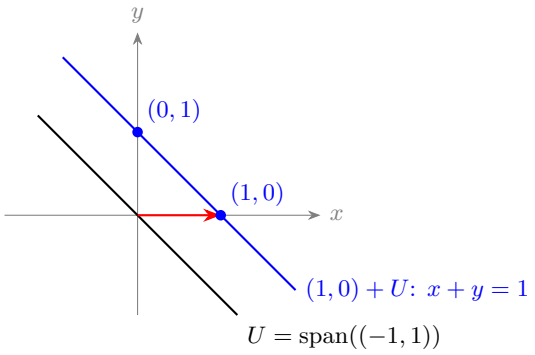
- The dimension of a product is the sum: $\dim(\mathbb{R}^2 \times \mathbb{R}^3) = 5$, not 6.
- For three or more subspaces, pairwise intersections equal to $\{0\}$ do not make the sum direct: the lines $\operatorname{span}(e_1)$, $\operatorname{span}(e_2)$, $\operatorname{span}(e_1 + e_2)$ in $\mathbb{R}^2$ meet pairwise only in 0, but the dimension of their sum is $2 < 1+1+1$.

§2.4 Affine subsets and quotient spaces (Ch. 8.2)

The question. Given a subspace U of V, how do we build a vector space in which vectors that differ by an element of U count as the same? In $\mathbb{R}^2$ with $U = \operatorname{span}((0,1))$, should $(1,2)$ and $(1,4)$ become equal, and what is their sum with $(2,1)$?

Definition (Def. 22, 23 / LADR 3.95, 3.97)**.** For $v \in V$ and a subspace U, $v + U = \{v + u : u \in U\}$. A subset of this form is an **affine subset**, **parallel** to U. Two affine subsets are parallel if they are parallel to the same subspace.

Examples 41 and 42. **(41)** The line $x + y = 1$ in $\mathbb{R}^2$ is $(1,0) + \operatorname{span}((-1,1))$: (x,y) is on the line exactly when $(x,y) - (1,0) = (x-1, y)$ has $(x-1) + y = 0$, that is, lies in $\operatorname{span}((-1,1))$.
(42) $\int x^2\,dx = \frac{1}{3}x^3 + C$ is the affine subset $\frac{1}{3}x^3 + \mathbb{R}$ of $\mathcal{P}(\mathbb{R})$, parallel to the constants.



$(1,0) + U$ is U moved by $(1,0)$ (red arrow). Any of its points can serve as the base point: it is also $(0,1) + U$.

Proposition (Prop. 38: an affine subset is parallel to only one subspace)
If $v+U = x+W$ for subspaces U, W, then $U = W$.

The idea

v lies in $x+W$, so $v = x+w_0$ with $w_0 \in W$; every $u \in U$ then shows up as a difference of two elements of W.

Proof. 1. $v = v+0 \in v+U = x+W$, so $v = x+w_0$ with $w_0 \in W$.

2. For $u \in U$, $v+u \in x+W$, so $v+u = x+w$ for some $w \in W$. Then $u = w-w_0 \in W$. So $U \subseteq W$.

3. Now start from $x \in x+W = v+U$: $x = v+u_0$ with $u_0 \in U$, and for $w \in W$, $x+w = v+u$ for some $u \in U$, so $w = u-u_0 \in U$. So $W \subseteq U$.

□

Proposition (Prop. 39 / LADR 3.101: when two parallel affine subsets are equal)
For a subspace U and $v, w \in V$ the following are equivalent:

$$\text{(i) } v-w \in U \qquad \text{(ii) } v+U = w+U \qquad \text{(iii) } (v+U)\cap(w+U) \neq \varnothing.$$

Proof. 1. **(i)⇒(ii).** For $u \in U$, $v+u = w+\big((v-w)+u\big) \in w+U$. So $v+U \subseteq w+U$. Since $w-v = -(v-w) \in U$, for $u \in U$ we have $w+u = v+\big((w-v)+u\big) \in v+U$, so $w+U \subseteq v+U$.

2. **(ii)⇒(iii).** $v = v+0$ lies in both.

3. **(iii)⇒(i).** If $v+u_1 = w+u_2$, then $v-w = u_2-u_1 \in U$.

□

So the affine subsets parallel to U either coincide or do not meet, and every vector of V lies in exactly one of them (namely $v \in v+U$).

Proposition (LADR 3E: solutions of a linear equation)
For $T \in \mathcal{L}(V,W)$ and $w \in W$, the solution set of $Tv = w$ is empty or equal to $v_0 + \operatorname{null} T$ for any one solution v_0.

Proof. If $Tv_0 = w$, then $Tv = w = Tv_0 \iff T(v-v_0) = 0 \iff v \in v_0 + \operatorname{null} T$. □

Instance: Ex. 41 is the solution set of $x+y=1$; there $v_0 = (1,0)$ and $\operatorname{null}\big((x,y)\mapsto x+y\big) = \operatorname{span}((-1,1))$.

Definition (Def. 24 / LADR 3.99, 3.102: quotient space)**.** V/U is the set of all affine subsets of V parallel to U, with

$$(v+U)+(w+U) = (v+w)+U, \qquad \lambda(v+U) = (\lambda v)+U.$$

The zero element is $0+U = U$.

The idea

An element of V/U is a whole set $v+U$, and the same set has many names: $v+U = v'+U$ whenever $v-v' \in U$. The operations are defined using a name, so there is something to check: if we rename both inputs, does the output set stay the same? This check is called *well-definedness*, and it is needed every time a map out of a quotient is defined by a formula in v. It is always the same computation: assume the differences of the names are in U, and show the difference of the outputs is in U. It works because U is closed under addition and scalar multiplication, which is why U must be a subspace.

Proof that V/U is a vector space. 1. Suppose $v_1+U = v_2+U$ and $w_1+U = w_2+U$. By Prop. 39, $v_1-v_2 \in U$ and $w_1-w_2 \in U$.

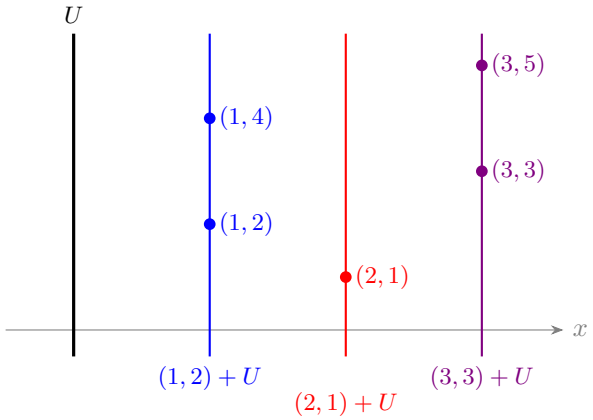
2. $(v_1+w_1)-(v_2+w_2) = (v_1-v_2)+(w_1-w_2) \in U$, so $(v_1+w_1)+U = (v_2+w_2)+U$.

3. $\lambda v_1 - \lambda v_2 = \lambda(v_1 - v_2) \in U$, so $\lambda v_1 + U = \lambda v_2 + U$.
4. **The axioms hold.** Each one follows by writing every element as $v + U$ and using the axiom in V. Associativity:

$$((u + U) + (v + U)) + (w + U) = ((u + v) + w) + U = (u + (v + w)) + U = (u + U) + ((v + U) + (w + U)).$$

The zero is $0 + U$, since $(v + U) + (0 + U) = v + U$. The additive inverse of $v + U$ is $(-v) + U$, since their sum is $0 + U$. The other axioms (commutativity, distributive laws, $1(v + U) = v + U$, $(ab)(v + U) = a(b(v + U))$) are checked the same way: each side equals $x + U$ for the vectors x given by the axiom in V.

□



The quotient $\mathbb{R}^2/U$ for $U = \text{span}((0, 1))$, the y-axis: each vertical line is one element of $\mathbb{R}^2/U$. $(1, 2) + U = (1, 4) + U$ is the same line, since $(1, 4) - (1, 2) \in U$. To add the blue and red lines, add any point of one to any point of the other: $(1, 2) + (2, 1) = (3, 3)$ and $(1, 4) + (2, 1) = (3, 5)$ both lie on the violet line $x = 3$.

The question answered. The vectors that count as the same as v form the affine subset $v + U$, and the set V/U of these subsets, with the operations of Def. 24, is a vector space. In $\mathbb{R}^2/\text{span}((0, 1))$, $(1, 2) + U = (1, 4) + U$ because $(1, 4) - (1, 2) = (0, 2) \in U$, and the sum with $(2, 1) + U$ is the line $(3, 3) + U$, whichever name is used.

Shortcuts

- **Equality in a quotient.** $v + U = w + U \iff v - w \in U$. Reason: Prop. 39. Every well-definedness check starts here. Instance: in $\mathbb{R}^3$ with $U = \text{span}((1, 1, 0))$, $(1, 2, 3) + U = (3, 4, 3) + U$, since the difference $(2, 2, 0)$ is in U.
- **Any point of an affine subset can serve as its base point.** If $a \in v + U$, then $a + U = v + U$. Reason: Prop. 39, (iii)⇒(ii). Instance: the line of Ex. 41 is $(1, 0) + \text{span}((-1, 1)) = (0, 1) + \text{span}((-1, 1))$.
- **Solutions of $Tv = w$.** The solution set is empty or $v_0 + \text{null}\, T$ for any one solution v_0. Reason: the proposition above. Instance: $x + y = 1$ gives $(1, 0) + \text{span}((-1, 1))$.
- **Is $v + U$ a subspace?** Only when $v \in U$, and then $v + U = U$. Reason: HW4 P6. Instance: $(0, 1) + \text{span}((0, 1)) = \text{span}((0, 1))$.

Traps

- An element of V/U is a set, and its name v is not determined by it: $(1, 2) + U = (1, 4) + U$ in the figure.
- A formula in v does not automatically define a map on V/U. For $U = \text{span}((0, 1))$, "$f((x, y) + U) = y$" is not a function, since $(1, 2) + U = (1, 4) + U$ would get the two values 2 and 4. "$g((x, y) + U) = x$" is well defined, since vectors that differ by an element of U have the same first coordinate.
- $v + U$ is not a subspace unless $v \in U$: the line $x + y = 1$ does not contain 0.

§3 Problem set

HW4 P1. Let $A = \left[\begin{smallmatrix} a & b \\ c & d \end{smallmatrix}\right]$. (a) Verify by direct multiplication that $A\left[\begin{smallmatrix} d & -b \\ -c & a \end{smallmatrix}\right] = (ad-bc)I$. (b) Define $\det A = ad - bc$. Prove that if $\det A \neq 0$ then A is invertible, and derive the formula for A^{-1}. (c) Prove, without any general determinant theory, that if $\det A = 0$ then A is not invertible.

Hint for (c): if A were invertible, multiply the identity in (a) by A^{-1}.

HW4 P2. Decide which of the matrices below are invertible (the second over $\mathbb{C}$, the third over $\mathbb{F}_5$), and find the inverse when it exists.

$$\begin{bmatrix} 2 & 1 \\ 5 & 3 \end{bmatrix}, \qquad \begin{bmatrix} 1 & i \\ i & -1 \end{bmatrix}, \qquad \begin{bmatrix} 1 & 2 \\ 3 & 1 \end{bmatrix}$$

Hint: compute $ad - bc$ in the given field.

HW4 P3 (LADR 3D, Exercise 3). Let V be finite-dimensional and $T\colon V \to V$ linear. Prove that the following are equivalent: (a) T is an isomorphism; (b) for every basis $v_1, \dots, v_n$ of V, the list $Tv_1, \dots, Tv_n$ is a basis of V.

Hint: an injective map sends independent lists to independent lists; then count.

HW4 P4. Suppose BC is defined. (a) Prove $\operatorname{rank}(BC) \leq \min\{\operatorname{rank} B, \operatorname{rank} C\}$. (b) Give an example where the inequality is strict although B and C both have positive rank.

Hint: $\operatorname{range}(BC)$ is contained in $\operatorname{range} B$, and it is the image of $\operatorname{range} C$ under B.

HW4 P5. Let V, W be finite-dimensional. Prove $\dim(V \times W) = \dim V + \dim W$ by constructing a basis of $V \times W$ from bases of V and W.

Hint: $(v_i, 0)$ and $(0, w_j)$.

HW4 P6. Let U be a subspace of V and $v, v' \in V$. (a) Prove that $v + U$ is a subspace of V if and only if $v \in U$; in that case $v + U = U$. (b) Prove that $v + U = v' + U$ if and only if $v - v' \in U$.

Hint: a subspace contains 0.

HW4 P7 (optional). Suppose U is a subspace of a finite-dimensional V and $S\colon U \to V$ is injective. Prove that S extends to an invertible operator $T\colon V \to V$.

Hint: extend a basis of U and, separately, the images of that basis; send the new vectors to the new vectors.

R1 (Practice Midterm 1, Question 2: a left inverse). Let V, W be finite-dimensional and $T\colon V \to W$ injective. (a) For a basis $v_1, \dots, v_n$ of V, prove $Tv_1, \dots, Tv_n$ is independent in W. (b) Prove there is a linear $S\colon W \to V$ with $ST = I_V$.

Hint: (a) apply T to a relation and use $\operatorname{null} T = \{0\}$; (b) extend $Tv_1, \dots, Tv_n$ to a basis of W and define S on that basis (W3).

R2 (Midterm 1 review, Exercise 9). Let U, W be subspaces of a finite-dimensional V with $U \cap W = \{0\}$. Show there is an isomorphism $U \times W \to U \oplus W$.

Hint: the map $\Gamma(u, w) = u + w$ of Prop. 37.

R3 (Midterm 1 review, Exercise 10). On the space $\mathbb{F}^\infty$ of all sequences $(a_1, a_2, \dots)$ (Week 1, Ex. 6) let $R(a_1, a_2, \dots) = (0, a_1, a_2, \dots)$. (i) Show R is injective but not surjective. (ii) Explain why no operator on a finite-dimensional space has this property.

Hint: (i) look at the first entry of $R(a)$; (ii) Prop. 31.

R4 (Midterm 1 review, Exercise 13). Let $P\colon V \to V$ be linear with $P^2 = P$. Show $V = \operatorname{null} P \oplus \operatorname{range} P$.

Hint: write $v = (v - Pv) + Pv$; for the intersection, write $v = Pw$ and apply P.

R5 (Midterm 1 review, Exercise 12). Let $D\colon \mathcal{P}_3(\mathbb{R}) \to \mathcal{P}_2(\mathbb{R})$ be differentiation. Find bases β of $\mathcal{P}_3(\mathbb{R})$ and γ of $\mathcal{P}_2(\mathbb{R})$ such that

$$\mathcal{M}(D) = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \end{bmatrix}.$$

Hint: column j says $D\beta_j = \gamma_j$ for $j \leq 3$ and $D\beta_4 = 0$; put a basis of $\operatorname{null} D$ last.

R6 (Midterm 1 review, Exercise 5). Prove that if V and W are isomorphic finite-dimensional spaces, then $\dim V = \dim W$.

Hint: rank–nullity for an isomorphism (Prop. 30).

R7 (Midterm 1 review, Exercise 8). Let $T \in \mathcal{L}(V)$, V finite-dimensional, and $v_1, \ldots, v_n$ a basis. Prove that T is an isomorphism if and only if $Tv_1, \ldots, Tv_n$ is linearly independent.

Hint: independent images of a basis give $\operatorname{null} T = \{0\}$; equal dimensions (Prop. 31).